

# M^0 Audit Summary

Prepared by Cyfrin

Version 2.0


**Lead Auditors**

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Immeas


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# 1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

# 2 About This Document

This document is a public summary of the security audit Cyfrin performed for M^0 Foundation on M^0. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated November 26, 2024, was delivered to M^0 Foundation, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

# 3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

# 4 Copyright and License

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# 5 Audit Overview

Over the course of 7 days, the Cyfrin team conducted an audit on the M^0 code provided by M^0 Foundation. The findings consist of 2 Medium severity issues with the remainder being informational.

**Summary**

| | |
|---|---|
| Project Name | M^0 |
| Repository | m-portal |
| Commit | 8f909076f4fc... |
| Audit Timeline | Oct 31st - Nov 8th, 2024 |
| Methods | Manual Review, Stateful Fuzzing |

**Issues Found**

| Severity | Found | Resolved | Acknowledged |
|---|---|---|---|
| Critical Risk | 0 | 0 | 0 |
| High Risk | 0 | 0 | 0 |
| Medium Risk | 2 | 2 | 0 |
| Low Risk | 0 | 0 | 0 |
| Informational | 1 | 1 | 0 |
| Gas Optimizations | 0 | 0 | 0 |
| Total | 3 | 3 | 0 |

# 6 Audit Scope

Following files were in the scope of the current audit

```
src/Portal.sol
src/HubPortal.sol
src/SpokePortal.sol
src/SpokeVault.sol
src/governance/Governor.sol
src/governance/Migrator.sol
src/governance/Configurator.sol
src/libs/PayloadEncoder.sol
src/libs/RegistrarReader.sol
src/libs/TypeConverter.sol
```

# 7 Risk Classification

| | **Impact: High** | **Impact: Medium** | **Impact: Low** |
|---|---|---|---|
| **Likelihood: High** | Critical | High | Medium |
| **Likelihood: Medium** | High | Medium | Low |
| **Likelihood: Low** | Medium | Low | Low |

# 8 Protocol Summary

M^0 Labs has developed a cross-chain token bridging protocol that enables users to transfer M tokens between different EVM-compatible blockchains using Wormhole as the messaging layer. The protocol follows a Hub-and-Spoke architecture:

- **Hub Portal:** deployed on Ethereum mainnet, serves as the central bridging point for M tokens.
- **Spoke Portals:** deployed on various L2s (Base and Optimism currently), handle bridging on their respective chains.
- **Spoke Vault:** bridges excess M tokens on L2s back to the hub.
- **Wormhole NttManager and Transceivers:** the protocol uses Wormhole's Native Token Transfer (NTT) Manager and Transceivers for cross-chain messaging.

# 9 Attestation

Cyfrin attests that it performed a security review of the M^0 code provided by M^0 Foundation.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-11-26-cyfrin-m0-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to M^0 Foundation has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=a598e9a2dac915c58e4be926c836a9b607e64f26655707d9327c4135740c75fe
F=2024-11-26-cyfrin-m0-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.